Chapter 7 Estimating Parameters and Determining Sample Sizes
Estimating a Population Standard Deviation or Variance (7.3)

Objectives

In this lesson, we will learn how to

- use a sample variance or sample standard deviation as a point estimate;
- understand the main features of the chi-square distribution;
- find left and right chi-square critical values;
- construct confidence intervals for a population variance σ^2;
- construct confidence intervals for a population standard deviation σ;
- determine sample sizes needed to estimate σ.

1 Main Idea

In earlier lessons, we estimated population proportions and population means. In this lesson, we estimate the amount of variation in a population.

The main population parameters are:

$$\sigma = \text{population standard deviation}$$

and

$$\sigma^2 = \text{population variance}.$$

Point Estimates

- The sample standard deviation s is used to estimate the population standard deviation σ.
- The sample variance s^2 is the best point estimate of the population variance σ^2.

2 The Chi-Square Distribution

The confidence interval methods in this section use the chi-square distribution, written as

$$\chi^2.$$

If the population is normally distributed, then the statistic

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

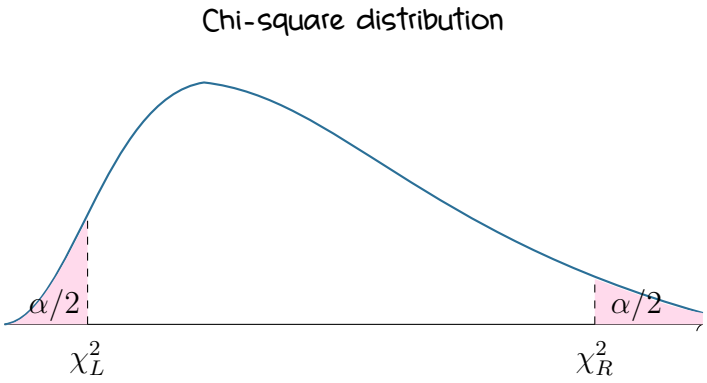
has a chi-square distribution with

$$df = n - 1$$

degrees of freedom.

Important Features of the Chi-Square Distribution

- The chi-square distribution is skewed to the right.
- Chi-square values are always 0 or positive; they are never negative.
- The shape depends on the degrees of freedom.
- As the degrees of freedom increase, the chi-square distribution becomes more like a normal distribution.



Important Caution

Because the chi-square distribution is not symmetric, a confidence interval for σ^2 does not have the form

$$s^2 - E < \sigma^2 < s^2 + E.$$

Instead, we must find separate lower and upper limits.

3 Left and Right Chi-Square Critical Values

For a confidence interval, we need two chi-square critical values:

$$\chi^2_L = \text{left-tailed chi-square critical value}$$

and

$$\chi^2_R = \text{right-tailed chi-square critical value.}$$

Table A-4 Caution

Table A-4 for the chi-square distribution gives areas to the right of the critical value.
This is different from the standard normal table, which gives cumulative areas from the left.

Example 1: Finding Chi-Square Critical Values: A simple random sample of 22 pulse rates is obtained. For a 95% confidence interval, find the left and right chi-square critical values.

Solution

The sample size is

$$n = 22.$$

So the degrees of freedom are

$$df = n - 1 = 22 - 1 = 21.$$

For a 95% confidence interval,

$$\alpha = 0.05, \qquad \alpha/2 = 0.025.$$

Using Table A-4 with $df = 21$:

$$\chi^2_R = 35.479.$$

For the left critical value, Table A-4 uses areas to the right, so we use

$$1 - 0.025 = 0.975.$$

Thus,

$$\chi^2_L = 10.283.$$

Therefore,

$$\boxed{\chi^2_L = 10.283 \quad \text{and} \quad \chi^2_R = 35.479.}$$

4 Confidence Interval for σ^2 and σ

Requirements

To construct a confidence interval for σ or σ^2:

1. The sample must be a simple random sample.
2. The population must have normally distributed values.

The normality requirement is very important in this section. Large departures from normality can lead to large errors.

Confidence Interval for the Population Variance σ^2

$$\frac{(n-1)s^2}{\chi_R^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_L^2}$$

Confidence Interval for the Population Standard Deviation σ

Take the square root of each part of the variance interval:

$$\sqrt{\frac{(n-1)s^2}{\chi_R^2}} < \sigma < \sqrt{\frac{(n-1)s^2}{\chi_L^2}}.$$

Round-Off Rule

- If original data are given, round the interval limits to one more decimal place than the original data.
- If summary statistics n and s are given, round the interval limits to the same number of decimal places used for s.

5 Procedure for Constructing a Confidence Interval

Steps

1. Check that the sample is simple random and that the population is normally distributed.
2. Compute degrees of freedom:
$$df = n - 1.$$
3. Find the critical values χ^2_L and χ^2_R.
4. Use the formula for the confidence interval for σ^2.
5. If the problem asks for σ, take square roots of the variance limits.
6. Round the final interval limits correctly.

Example 2: Confidence Interval for σ: Listed below is a simple random sample of pulse rates, in beats per minute, obtained from adult females.

76	76	86	74	66	62	78	68	62	62	74
80	54	74	74	84	60	52	84	66	56	66

Construct a 95% confidence interval estimate of the population standard deviation σ.

Solution

The sample size is
$$n = 22.$$
The sample standard deviation is
$$s = 9.91959.$$

For a 95% confidence interval,

$$\alpha = 0.05, \qquad \alpha/2 = 0.025.$$

The degrees of freedom are

$$df = n - 1 = 22 - 1 = 21.$$

From Table A-4,

$$\chi^2_L = 10.283$$

and

$$\chi^2_R = 35.479.$$

First construct the confidence interval for the variance:

$$\frac{(n-1)s^2}{\chi^2_R} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_L}.$$

Substitute the values:

$$\frac{(21)(9.91959)^2}{35.479} < \sigma^2 < \frac{(21)(9.91959)^2}{10.283}.$$

This gives

$$58.2419 < \sigma^2 < 200.9495.$$

Now take the square root of each part:

$$\sqrt{58.2419} < \sigma < \sqrt{200.9495}.$$

So

$$7.6 < \sigma < 14.2.$$

Therefore, the 95% confidence interval estimate of the population standard deviation is

$$\boxed{7.6 \text{ bpm} < \sigma < 14.2 \text{ bpm}}.$$

Interpretation

We are 95% confident that the interval from 7.6 bpm to 14.2 bpm contains the true value of the population standard deviation σ for adult female pulse rates.
This interval cannot be written as $s \pm E$.

6 Using Confidence Intervals for Comparisons

Confidence intervals can be used informally to compare variation in different data sets.

However, overlapping confidence intervals should not be used as a formal test for equality of variances or equality of standard deviations.

Key Idea

A confidence interval for σ gives a reasonable range of values for the population standard deviation, but comparing two intervals by eye is only informal.

7 Sample Size for Estimating σ

Finding the sample size needed to estimate a population standard deviation is more complicated than finding sample sizes for proportions or means.

For normally distributed populations, tables or technology can be used.

Sample Sizes for Estimating σ

Desired Accuracy	95% Confidence	99% Confidence
Within 1% of σ	19,205	33,218
Within 5% of σ	768	1,336
Within 10% of σ	192	336
Within 20% of σ	48	85
Within 30% of σ	21	38
Within 40% of σ	12	22
Within 50% of σ	8	14

Example 3: Sample Size for Estimating σ: We want to estimate the standard deviation σ of all pulse rates of adult women. We want to be 99% confident that our estimate is within 5% of the true value of σ. How large should the sample be? Assume that the population is normally distributed.

Solution

Using the table for estimating σ, for

$$99\% \text{ confidence}$$

and accuracy within

$$5\% \text{ of } \sigma,$$

the required sample size is

$$\boxed{1336}.$$

We should obtain a simple random sample of at least 1336 pulse rates of adult women.

Summary

Key Ideas

- The sample variance s^2 is the best point estimate of the population variance σ^2.
- The sample standard deviation s is used to estimate the population standard deviation σ.
- Confidence intervals for σ and σ^2 use the chi-square distribution.
- For this section, the degrees of freedom are

$$df = n - 1.$$

- The chi-square distribution is skewed right and is not symmetric.
- Table A-4 gives areas to the right of chi-square critical values.
- To find a confidence interval for σ, first find the interval for σ^2, then take square roots.
- The normality requirement is especially important for confidence intervals involving standard deviation or variance.